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First Name(s):														

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

June exam

5 June 2023

Time : 2 hours

Maximum marks: 60

Full marks: 60

- This question paper consists of 10 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators also allowed.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
8	13	9	4	6	5	15	60

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Question 1. [8 points] Let

$$A = \begin{pmatrix} 1 & 3 & 1 & -2 & 4 & 0 \\ 1 & 3 & 3 & 2 & 1 & 3 \\ 2 & 6 & 4 & 1 & 8 & 10 \\ 3 & 9 & 7 & 2 & 6 & 11 \\ 4 & 12 & 10 & 4 & 7 & 14 \end{pmatrix} \quad \text{and} \quad U = \begin{pmatrix} 1 & 3 & 1 & -2 & 4 & 0 \\ 0 & 0 & 2 & 4 & -3 & 3 \\ 0 & 0 & 0 & 1 & 3 & 7 \\ 0 & 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The matrix U is a row-echelon form of the matrix A . You need not check this.

Use this information to answer the following questions.

(a) Give a basis for (i) $RS(A)$ and (ii) $CS(A)$. You need not give reasons for your answers.

(i) A basis for $RS(A)$ is $\{(1, 3, 1, -2, 4, 0), (0, 0, 2, 4, -3, 3), (0, 0, 0, 1, 3, 7), (0, 0, 0, 0, 0, 5)\}$.

(ii) A basis for $CS(A)$ is $\left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 4 \\ 7 \\ 10 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \\ 1 \\ 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 10 \\ 11 \\ 14 \end{pmatrix} \right\}$.

(b) Give the rank and nullity of A .

$\text{Rank}(A) = 4$ and $\text{Nullity}(A) = 6 - 4 = 2$.

(c) Find the nullity of A^T .

$\text{Nullity}(A^T) = 5 - 4 = 1$.

(d) Is it true that for every $\mathbf{v} \in \mathbb{R}^5$ there is a $\mathbf{u} \in \mathbb{R}^6$ such that $\mathbf{v} = A\mathbf{u}$? Give a reason for your answer.

No.

$\dim(CS(A)) = 4$, while $\dim(\mathbb{R}^5) = 5$. Hence not every vector $\mathbf{v}$ in $\mathbb{R}^5$ can be written as a linear combination of columns in A .

(e) Is it possible to obtain an invertible matrix by deleting one of the columns of A ? If so, which column should be deleted, and why? If not, explain why not.

No, this is not possible.

The columns of any invertible matrix are linearly independent.

The dimension of the column space of A is 4. This implies that any 5 columns of A are linear dependent. Hence any matrix obtained by deleting one of the 6 columns of A will be not invertible.

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Question 2. [13 points] Let $A = \begin{pmatrix} 6 & 0 & 0 \\ 2 & 0 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.

(a) Find all the eigenvalues of A .

$$\begin{vmatrix} 6 - \lambda & 0 & 0 \\ 2 & -\lambda & 1 \\ 0 & 0 & 2 - \lambda \end{vmatrix} = 0.$$

We may for instance expand along row 1 and obtain

$$(6 - \lambda) \begin{vmatrix} -\lambda & 1 \\ 0 & 2 - \lambda \end{vmatrix} = 0,$$

$$(6 - \lambda)(-\lambda)(2 - \lambda) = 0.$$

Hence the eigenvalues are $\lambda = 0, 2, 6$.

(b) Can A be written as a product of elementary matrices? Give a reason for your answer, using only your answer to (a).

0 is an eigenvalue of A . Hence A is singular and can therefore not be written as a product of elementary matrices.

(c) Find elementary matrices E_1 and E_2 and a matrix U in row-echelon form such that $A = E_2 E_1 U$.

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 1/3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

(d) For each of the eigenvalues of A , find the eigenspace corresponding to that eigenvalue.

$$\begin{aligned} W_0 &= NS \begin{pmatrix} 6 & 0 & 0 \\ 2 & 0 & 1 \\ 0 & 0 & 2 \end{pmatrix} = \left\{ s \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} : s \in \mathbb{R} \right\}, \\ W_2 &= NS \begin{pmatrix} 4 & 0 & 0 \\ 2 & -2 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \left\{ s \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} : s \in \mathbb{R} \right\}, \\ W_6 &= NS \begin{pmatrix} 0 & 0 & 0 \\ 2 & -6 & 1 \\ 0 & 0 & -4 \end{pmatrix} = \left\{ s \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} : s \in \mathbb{R} \right\}. \end{aligned}$$

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- (e) Is there an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix D ? If so, give P and D ; if not, say why not.

Yes, we can for instance take

$$P = \begin{pmatrix} 0 & 0 & 3 \\ 1 & 1 & 1 \\ 0 & 2 & 0 \end{pmatrix} \text{ and } D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{pmatrix}.$$

- (f) For this matrix A , does there exist a positive integer k such that $A^k = \mathbf{0}$ (the zero matrix)? Justify your answer.

Suppose $A^k = \mathbf{0}$. Since $A = PDP^{-1}$, this implies

$$PD^kP^{-1} = \mathbf{0}.$$

Multiplying both sides of the equation by P^{-1} on the left and by P to the right, we obtain

$$D^k = \mathbf{0},$$

which is a contradiction because

$$D^k = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2^k & 0 \\ 0 & 0 & 6^k \end{pmatrix}.$$

Question 3. [9 points]

Let $T : M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^4$ be a linear transformation such that

$$\begin{aligned} T\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right) &= \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}, & T\left(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right) &= \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \\ T\left(\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\right) &= \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}, & T\left(\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right) &= \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}. \end{aligned}$$

- (a) Find $T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)$.

$$T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = a \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} + c \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} + d \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -b+c \\ -a+d \\ a-d \\ b-c \end{pmatrix}.$$

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- (b) Find $\ker(T)$ and a basis for $\ker(T)$.

To find the kernel, we solve the system of equations

$$-b + c = 0$$

$$-a + d = 0.$$

We obtain

$$\ker(T) = \left\{ \begin{pmatrix} t & s \\ s & t \end{pmatrix} : s, t \in \mathbb{R} \right\}.$$

A basis for $\ker(T)$ is for instance

$$\left\{ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

- (c) Find $\text{ran}(T)$ and $\dim(\text{ran}(T))$.

$$\text{ran}(T) = \left\{ \begin{pmatrix} s \\ t \\ -t \\ -s \end{pmatrix} : s, t \in \mathbb{R} \right\}.$$

$$\dim(\text{ran}(T)) = 2.$$

- (d) Is T injective? Give a reason for your answer.

T is not injective (for instance) because $\ker(T) \neq \{\mathbf{0}\}$.

- (e) Is T surjective? Give a reason for your answer.

T is not surjective (for instance) because $\dim(\text{ran}(T)) = 2 < 4 = \dim(\mathbb{R}^4)$.

- (f) Find the matrix ${}_C[T]_B$ representing T with respect to the ordered bases

$$B = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\} \text{ and } C = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

for $M_{2 \times 2}(\mathbb{R})$ and $\mathbb{R}^4$ respectively.

$${}_C[T]_B = \begin{pmatrix} 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{pmatrix}$$

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Question 4. [4 points] On P_2 , an inner product can be defined by

$$\langle a_2x^2 + a_1x + a_0, b_2x^2 + b_1x + b_0 \rangle = a_2b_2 + a_1b_1 + a_0b_0,$$

where $a_0, a_1, a_2, b_0, b_1, b_2 \in \mathbb{R}$. (You need not check that this defines an inner product.)

Let $p(x) = 3x - 4$ and $q(x) = 7x^2$.

(a) Find $\langle p(x), q(x) \rangle$.

$$\langle p(x), q(x) \rangle = \langle 3x - 4, 7x^2 \rangle = 0.$$

(b) Find $\|p(x)\|$.

$$\|p\| = \sqrt{\langle p, p \rangle} = \sqrt{\langle 3x - 4, 3x - 4 \rangle} = \sqrt{3^2 + 4^2} = 5.$$

(c) Let A be defined by

$$A = \{r \in P_2 : \langle r, x^2 - 2x + 4 \rangle = 0\}.$$

Find two linearly independent vectors in A .

We need $a_0, a_1, a_2 \in \mathbb{R}$ such that

$$\langle a_2x^2, a_1x + a_0, x^2 - 2x + 4 \rangle = 0,$$

which means

$$a_2 - 2a_1 + 4a_0 = 0.$$

We can for instance take

$$a_0 = 0, a_1 = 1, a_2 = 2$$

and

$$a_0 = 1, a_1 = 0, a_2 = -4$$

From this, we obtain the linearly independent vectors $2x^2 + x$ and $-4x^2 + 1$.

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Question 5. [6 points] Let S be the subset of P_3 defined by

$$S = \{4x^3 + x^2, -x^2 + 2x, 2x^3 + x, -3x^2 + 1\}.$$

Answer the following questions, giving reasons for your answers.

(a) Find a basis B for $\text{span}(S)$.

The set S leads to the matrix

$$\begin{pmatrix} 4 & 0 & 2 & 0 \\ 1 & -1 & 0 & -3 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

which is row equivalent to the matrix

$$\begin{pmatrix} 4 & 0 & 2 & 0 \\ 0 & -1 & -1/2 & -3 \\ 0 & 0 & 0 & -6 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

in row echelon form. The pivot columns in this matrix tell us that we get a basis for $\text{span}(S)$ if we take the first, the second, and the fourth polynomial in S , i.e.,

$$B = \{4x^3 + x^2, -x^2 + 2x, -3x^2 + 1\}$$

is a basis for $\text{span}(S)$.

(b) What is $\dim(\text{span}(S))$?

From the above, $\dim(\text{span}(S)) = 3$.

(c) Is $x^3 + x^2 \in \text{span}(S)$?

The question leads to a linear system corresponding to the augmented matrix

$$\left(\begin{array}{cccc|c} 4 & 0 & 2 & 0 & 1 \\ 1 & -1 & 0 & -3 & 1 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right),$$

which has no solution.

Hence $x^3 + x^2 \notin \text{span}(S)$.

(d) Is there a $p(x) \in P_3$ such that $B \cup \{p(x)\}$ is a basis for P_3 ?

We can for instance take $p(x) = x^3 + x^2$.

Question 6. [5 points] Let $S = \{A \in M_{4 \times 5}(\mathbb{R}) : \text{Rank}(A) \leq 1\}$.

Prove or disprove each of the following statements:

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(a) S is closed under scalar multiplication.

S is closed under scalar multiplication:

Let $A \in S$ and λ any scalar. Then $\text{Rank}(A) \leq 1$, which means $\dim(CS(A)) \leq 1$.

Suppose $\lambda \neq 0$. Then $CS(A) = CS(\lambda A)$. It follows that $\dim(CS(\lambda A)) \leq 1$, and therefore, $\text{Rank}(\lambda A) \leq 1$. Hence $\lambda A \in S$.

On the other hand, if $\lambda = 0$ then $\text{Rank}(\lambda A) = 0$ and hence also $\lambda A \in S$.

(b) S is a subspace of $M_{4 \times 5}(\mathbb{R})$.

S is not closed under addition, and hence S is not a subspace of $M_{4 \times 5}(\mathbb{R})$:

Let for instance

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Then $\text{Rank}(A) = \text{Rank}(B) = 1$ and $\text{Rank}(A + B) = 2$. Hence $A, B \in S$ but $A + B \notin S$.

Question 7. [15 points] Read the argument below and then answer the questions that follow.

Proof: Let V and W be vector spaces, $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ be a basis for V , $T : V \rightarrow W$ be a linear transformation and $C = \{T(\mathbf{b}) : \mathbf{b} \in B\}$. If $\mathbf{w} \in \text{ran}(T)$, then there is a $\mathbf{v} \in V$ such that $\mathbf{w} = T(\mathbf{v})$ (1)

We can find scalars $\lambda_1, \lambda_2, \dots, \lambda_k$ such that

$$\mathbf{v} = \lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k. \dots\dots\dots (2)$$

Then

$$\mathbf{w} = T(\mathbf{v}) = \lambda_1 T(\mathbf{b}_1) + \lambda_2 T(\mathbf{b}_2) + \dots + \lambda_k T(\mathbf{b}_k). \dots\dots\dots (3)$$

$$\text{It follows that } \text{ran}(T) \subseteq \text{span}(C). \dots\dots\dots (4)$$

$$\text{Since we also have } \text{span}(C) \subseteq \text{ran}(T), \dots\dots\dots (5)$$

it follows that $\text{span}(C) = \text{ran}(T)$

(a) Carefully state the result proved here.

Let V and W be vector spaces, $B = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ be a basis for V and $T : V \rightarrow W$ be a linear transformation. Then $\{T(\mathbf{b}) : \mathbf{b} \in B\}$ spans $\text{ran}(T)$.

(b) Why, at (1), can $\mathbf{v} \in V$ be found such that $\mathbf{w} = T(\mathbf{v})$?

Because $\mathbf{w}$ is in the range of T .

(c) Why can scalars $\lambda_1, \lambda_2, \dots, \lambda_k$ be found such that (2) holds?

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Because $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ is a basis for V

(d) Where and how is the fact that T is linear used in this proof?

At (3) where we conclude that $T(\mathbf{v}) = \lambda_1 T(\mathbf{b}_1) + \lambda_2 T(\mathbf{b}_2) + \dots + \lambda_k T(\mathbf{b}_k)$.

(e) Explain how (4) follows from (3).

In (3), we have shown that every vector $\mathbf{w}$ in the range of T can be written as a linear combination of the vectors in C . Hence the range of T is in the span of C .

(f) Let V, W and T be as in the proof, A a finite subset of V and

$$D = \{T(\mathbf{v}) : \mathbf{v} \in A\}.$$

For each of the following statements, either prove it, or disprove it by giving a counter example:

(i) If A spans V , then D spans $\text{ran}(T)$.

This is true.

Let $A = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a subset of V such that A spans V . If $\mathbf{w} \in \text{ran}(T)$, then there is a $\mathbf{v} \in V$ such that $\mathbf{w} = T(\mathbf{v})$. Since A spans V , we can find scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ such that

$$\mathbf{v} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n.$$

Then

$$\mathbf{w} = T(\mathbf{v}) = \lambda_1 T(\mathbf{v}_1) + \lambda_2 T(\mathbf{v}_2) + \dots + \lambda_n T(\mathbf{v}_n).$$

It follows that

$$\text{ran}(T) \subseteq \text{span}\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\} = \text{span}(D).$$

Because $D \subseteq \text{ran}(T)$ and $\text{ran}(T)$ is a subspace of W , we also have that

$$\text{span}(D) \subseteq \text{ran}(T).$$

Therefore, $\text{span}(D) = \text{ran}(T)$

(ii) If A is a linearly dependent subset of V , then D is a linearly dependent subset of W .

This is true.

Let $A = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a linearly dependent subset of V . Then there exist scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ not all zero such that

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0}.$$

It follows that

$$T(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n) = T(\mathbf{0}) = \mathbf{0},$$

and hence

$$\lambda_1 T(\mathbf{v}_1) + \lambda_2 T(\mathbf{v}_2) + \dots + \lambda_n T(\mathbf{v}_n) = \mathbf{0},$$

where not all λ_i are equal to zero. Hence $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly dependent.

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- (iii) If A is a linearly independent subset of V , D is a linearly independent subset of W .

This is false.

Let for instance $V = W = \mathbb{R}^2$ and T the linear transformation such that for every $\mathbf{v} \in V$, $T(\mathbf{v}) = \mathbf{0}$.

Further, let

$$A = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

Then

$$D = \{\mathbf{0}\},$$

which is not linearly independent.

- (iv) If A is a basis for V , then D is a basis for $\text{ran}(T)$.

This is false.

Let for instance $V = W = \mathbb{R}^2$ and T the linear transformation defined by

$$T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

Further, let

$$A = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}.$$

Then

$$D = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right\},$$

which is not linearly independent.